

MCALLEN, TX, USA

WMO: 722506

Lat: 26.176N

Lon: 98.240W

Elev: 100

StdP: 14.64

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 38.5	(c) 42.0	(d) 18.7	(e) 14.1	(f) 56.4	(g) 24.3	(h) 18.5	(i) 56.3	(j) 27.0	(k) 73.4	(l) 25.0	(m) 73.0	(n) 9.0	(o) 340	(p) 0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 19.7	(c) 101.4	(d) 76.4	(e) 99.7	(f) 76.8	(g) 98.1	(h) 76.7	(i) 80.4	(j) 91.3	(k) 79.9	(l) 90.3	(m) 79.3	(n) 89.1	(o) 14.3	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.6	149.0	82.6	77.8	144.9	82.6	77.2	142.0	82.5	44.1	91.4	43.4	89.4	42.8	88.8	92.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 25.0	(b) 22.9	(c) 20.5	(e) 33.5	(f) 104.6	(g) 3.6	(h) 2.9	(i) 30.9	(j) 106.7	(k) 28.8	(l) 108.4	(m) 26.8	(n) 110.0	(o) 24.1	(p) 112.1		
(a) 30.2	(b) 82.5	(c) 2.9	(e) 2.3	(f) 28.1	(g) 84.1	(h) 26.3	(i) 85.5	(j) 24.7	(k) 86.7	(l) 22.5	(m) 88.4	(n) 88.4	(o) 88.4	(p) 88.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	76.4	62.0	66.2	71.3	77.3	82.6	86.5	87.5	88.1	83.9	77.8	69.7	63.0
	DBStd	11.43	9.30	9.74	8.05	6.23	4.50	2.87	2.98	3.03	4.01	6.61	8.35	9.31
	HDD50	40	18	8	1	0	0	0	0	0	0	0	2	11
	HDD65	517	168	97	38	5	0	0	0	0	0	6	50	153
	CDD50	9666	389	463	663	818	1012	1094	1163	1180	1016	862	593	413
	CDD65	4667	74	132	234	373	547	644	698	715	566	403	191	90
	CDH74	51361	479	922	1711	3326	6027	8120	9052	9347	6399	3930	1455	593
	CDH80	25003	103	302	606	1374	2843	4327	4975	5195	3047	1642	451	138
Wind	WSAvg	10.5	9.6	10.8	11.5	12.3	12.6	11.8	12.1	10.6	8.1	8.8	9.0	9.0
Precipitation	PrecAvg	23.0	1.3	1.2	0.7	1.4	2.9	2.8	1.6	2.4	4.3	2.4	1.1	1.1
	PrecMax	35.7	4.6	5.2	3.8	5.9	7.9	10.5	8.3	8.1	10.9	7.5	4.2	3.9
	PrecMin	8.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.0	0.0	0.0
	PrecStd	6.5	1.3	1.1	1.0	1.5	2.2	2.6	2.0	2.0	2.8	2.0	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.9	92.5	94.8	97.3	99.2	101.7	103.2	103.5	101.4	96.5	91.7	87.7
		MCWB	68.3	68.9	69.5	73.8	77.9	77.6	76.3	76.8	75.2	76.3	74.5	70.9
	2%	DB	82.7	87.4	89.4	93.4	96.3	99.1	100.8	101.5	98.0	93.4	88.0	83.9
		MCWB	67.6	69.9	71.8	73.7	77.4	77.6	76.7	76.8	76.3	75.4	72.2	69.6
	5%	DB	79.6	83.7	86.3	90.7	94.3	97.5	99.1	99.8	95.4	91.1	85.2	80.6
		MCWB	66.1	69.1	70.4	73.6	76.7	77.1	76.8	76.7	76.3	74.8	70.9	68.6
	10%	DB	75.7	79.9	83.1	87.9	92.2	95.7	97.2	97.8	92.9	88.7	82.2	76.9
		MCWB	64.9	67.1	68.9	72.4	75.9	76.8	76.6	76.7	76.2	74.2	69.9	67.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	74.2	75.4	79.2	81.2	81.4	80.6	80.5	80.7	80.4	77.6	74.3
		MCDB	80.3	84.8	87.0	91.0	94.8	94.8	91.9	91.3	88.9	87.5	85.5	81.8
	2%	WB	70.6	72.7	74.0	77.3	79.7	80.4	79.8	79.9	79.9	79.2	75.6	72.8
		MCDB	77.9	81.3	83.8	88.2	91.9	91.9	90.5	90.5	88.1	86.2	82.4	79.5
	5%	WB	69.2	71.3	72.8	75.9	78.5	79.7	79.1	79.2	79.2	78.2	74.1	71.3
		MCDB	75.7	78.9	81.3	85.5	89.0	90.0	89.4	89.6	87.4	85.3	80.7	77.4
	10%	WB	67.7	69.9	71.6	74.9	77.7	78.9	78.4	78.5	78.4	77.1	72.7	69.6
		MCDB	73.5	76.6	79.0	83.4	87.2	88.7	88.4	88.6	86.7	84.3	79.2	75.4
Mean Daily Temperature Range	5% DB	MDBR	19.9	19.9	19.5	19.4	18.4	19.1	19.4	19.7	18.0	19.2	19.4	19.1
		MCDBR	23.1	22.9	21.3	21.2	20.2	20.8	21.5	21.8	20.8	20.2	20.9	21.7
		MCWBR	11.2	9.1	7.7	7.4	5.4	4.2	4.3	4.4	4.7	6.4	8.0	10.7
	5% WB	MCDBR	19.1	19.1	18.2	18.3	17.7	18.3	18.6	19.2	16.9	16.7	17.6	18.0
MCWBR		10.1	8.8	7.6	7.0	5.6	4.5	4.4	4.5	4.8	6.6	8.6	10.2	
Clear-Sky Solar Irradiance	taub	0.344	0.363	0.390	0.412	0.414	0.407	0.419	0.411	0.416	0.376	0.366	0.344	
	taud	2.494	2.430	2.358	2.310	2.343	2.415	2.384	2.418	2.373	2.479	2.455	2.500	
	Ebn at Noon	288	290	287	282	279	279	276	279	275	282	278	283	
	Edh at Noon	30	35	39	42	41	37	39	37	38	32	31	29	
All-Sky Solar Radiation	RadAvg	1011	1232	1501	1764	1909	2136	2121	2034	1683	1464	1136	917	
	RadStd	113	187	177	163	135	80	151	94	130	145	126	132	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.85	N/A	N/A	+1.12	N/A	N/A	N/A	N/A	+356	+295
(47) Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+171

Nomenclature: See separate page